

SMART Design for Multi-Turn Medical Agents

我们提出 **SMART (Sequential Multiple Assignment Randomized Trial) for medical LLM agents**, 在对话的多个 decision point 上 re-randomize agent 的 policy, identify 最优 **dynamic treatment regime**, 把 medical chatbot 的评估从"这个 agent 整体有没有用"升级到"在对话的哪一步、对哪类病人、该用哪种策略"。

We introduce a sequential randomized evaluation framework for medical LLM agents that identifies which conversational policy works best, for whom, and at which decision point.

Why This Matters

现状的 gap:

- 所有 medical LLM agent 的 evaluation (Nature Medicine 的 PreA trial、PROSCA、MedAgentBench、AgentClinic、MATRIX 等) 都把 agent 当成单一不可分割的 **intervention**: enrollment 时 randomize 一次, 结束时测 outcome。只能回答"agent 整体 vs. 无 agent", 回答不了"对话中哪个 sub-decision 真正驱动了 outcome"。
- 但 medical conversation 本质是 **sequential decision process**——每一轮都是一个 policy choice (问什么、何时 retrieve、何时 escalate), 下一步的最优 action 还依赖于前一步病人的 response。single-stage RCT 估出来的 ATE 把这整条 decision chain 平均掉了, 学不到 adaptive policy。
- 现有 multi-turn 评估 (MediQ, DoctorAgent-RL, AgentClinic) 虽然承认 sequential 结构, 但停留在"让 agent 自己决定怎么对话", 没有在 **decision point** 上做 **randomization**, 因此只能 observe policy、无法 identify optimal policy。

以前为什么没人做:

- 行为医学/精准医学领域 (Murphy 2005 以来) 的 SMART 和 dynamic treatment regime 方法论很成熟, 但他们的 "treatment" 是药物 / 行为干预, action space 是 discrete 的几个选项, decision point 是固定的几次随访。他们完全不碰 **LLM agent**。
- **LLM agent** 评估领域 (ML / NLP 社区) 会做 ablation、会跑 benchmark, 但对 **sequential randomization + causal identification of DTR** 这套统计工具几乎没有认知。现有 RL-based 方法 (DoctorAgent-RL) 能学 policy, 但给不出 finite-sample、trial-based、带 CI 的 unbiased value estimate。
- 两边没有交叉过, medical conversational agent 的 evaluation 正好卡在中间。

我们的贡献: 第一个把 SMART 的多阶段 **randomization + DTR identification theory** 搬到 medical LLM agent 上, 扩展到 conversational、high-dimensional history 的 action/state space, 让 medical chatbot 的评估从"测整体准确率"升级到"identify optimal adaptive policy with statistical guarantees"——并配套开源一个可复用的 SMART protocol + simulation infrastructure, 让社区能在新 model、新 scaffold 上直接跑 trial。

1. One-Sentence Pitch

We introduce the first **Sequential Multiple Assignment Randomized Trial (SMART)** framework for medical LLM agents, which re-randomizes conversational policy at multiple decision points to identify the **optimal dynamic treatment regime**—moving evaluation from “*does the chatbot work?*” to “*which sequence of conversational strategies works best, for whom, and when?*”

2. Why This Matters

- **The gap.** Every existing evaluation of medical LLM agents (Nature Medicine's PreA 2026, MedAgentBench, AgentClinic, MATRIX, MedDialogRubrics) treats the agent as a single monolithic intervention: randomize once, measure outcome at the end. This cannot tell us *which sub-decisions within a conversation drive outcomes*.
- **The insight.** A medical conversation is a sequential decision process. The right action at turn t depends on turns $1 \dots t-1$. SMART designs (Murphy, 2005) are the statistically principled tool for this, yet they have **zero prior application to LLM evaluation**.
- **Why now.** Patient-simulator infrastructure (MediQ, AgentClinic, PATIENT- Ψ) has matured; real digital-health RCTs are appearing (PreA, PROSCA); and ICLR 2026 just gave an Oral to MedAgentGym, showing the community rewards rigorous eval infrastructure for medical agents.

3. Project Setting

- **Deliverable:** (a) a SMART protocol + open codebase, (b) a theoretical identification story, (c) empirical results on a simulator + semi-synthetic data.
- **Core claim:** We can identify the *optimal dynamic treatment regime* for medical LLM agents with finite-sample guarantees, and doing so reveals policy choices that monolithic RCT-style evaluation misses.

4. Technical Setup (High Level)

Three decision points in the conversation. At each, the agent is randomized to one of several policy arms:

Stage	Decision	Example arms
Stage 1	Info gathering: how to gather history	(a) comprehensive ROS, (b) chief-complaint-focused, (c) red-flag-first

Stage 2	Tool use: when/whether to retrieve	(a) always retrieve guidelines, (b) conditional on uncertainty, (c) parametric only
Stage 3	Output: how to conclude	(a) single diagnosis + plan, (b) differential + shared decision, (c) escalate to human

Re-randomization rule: after each stage, compute an intermediate response score R_t (rubric-based, e.g. "information sufficiency"). Re-randomize conditional on R_t . This is what makes it SMART vs. factorial.

Patient simulator: built on MedQA / NEJM cases, following AgentClinic / MediQ patterns with an anti-hallucination patient agent.

Agent under test: 3–5 base LLMs × stage-arm scaffolds (implemented via LangGraph or a simple state machine).

Analysis: Q-learning (backward induction) + A-learning (doubly-robust) to estimate the optimal DTR and its value. Bootstrap for CIs.

4. Reading List

Read in this order. ★ = *must-read before first meeting*.

A. SMART methodology & dynamic treatment regimes (the statistics foundation)

- ★ **Murphy (2005)** — original SMART paper. *Statistics in Medicine*.
<https://pubmed.ncbi.nlm.nih.gov/15586449/> - Zihan Xia
<https://onlinelibrary.wiley.com/doi/10.1002/sim.2022>
- ★ **Almirall, Nahum-Shani, Sherwood, Murphy (2014)** — "Introduction to SMART designs." Best accessible intro. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4167891/>
- Nahum-Shani et al. (2023)** — "Sequential, Multiple Assignment, Randomized Trial Designs" (JAMA primer). <https://pmc.ncbi.nlm.nih.gov/articles/PMC10061579/>
- ★ **Schulte, Tsiatis, Laber, Davidian (2014)** — "Q- and A-learning Methods for Estimating Optimal Dynamic Treatment Regimes." *Statistical Science*.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC4300556/> - Jiawei Li
- Chakraborty & Moodie (2013)** — *Statistical Methods for Dynamic Treatment Regimes* (textbook, chapter on Q-learning, SMART).
- Chakraborty & Murphy (2014)** — "Dynamic Treatment Regimes" review. *Annual Review of Statistics*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4231831/>
- Kidwell et al. (2018)** — design + analysis of binary-outcome SMARTs (useful for our diagnostic-accuracy endpoint). <https://doi.org/10.1080/02664763.2017.1386773>

8. **Lei et al. (2012)** — "A 'SMART' design for building individualized treatment sequences." *Annu Rev Clin Psychol*.

B. Medical LLM agent evaluation (what we're differentiating from)

9. ★ **Li et al. (2024)** — **MediQ**: question-asking LLMs for interactive clinical reasoning. NeurIPS 2024. *Closest prior work*. <https://arxiv.org/abs/2406.00922> • Code: <https://github.com/stellalisy/MediQ> Ray Zhang
10. ★ **Schmidgall et al. (2024)** — **AgentClinic**: multimodal agent benchmark in simulated clinical environments. <https://arxiv.org/abs/2405.07960> • <https://agentclinic.github.io/>
11. **Feng et al. (2025)** — **DoctorAgent-RL**: RL-based multi-turn clinical dialogue. Directly relevant for framing against pure RL. <https://arxiv.org/abs/2505.19630> • Code: <https://github.com/JarvisUSTC/DoctorAgent-RL>
12. **Xu et al. (2025)** — **MedAgentGym** (ICLR 2026 Oral): code-centric biomedical reasoning training environment. <https://arxiv.org/abs/2506.04405> • Code: <https://github.com/wshi83/MedAgentGym>
13. **Jiang et al. (2025)** — **MedAgentBench**: FHIR-compliant EHR environment. <https://arxiv.org/abs/2501.14654>
14. **MATRIX (2025)** — safety-oriented simulation framework for conversational clinical agents. <https://arxiv.org/abs/2508.19163>
15. **Tang et al. (2025)** — **MedAgentsBench**: benchmarking thinking models & agent frameworks for complex medical reasoning. <https://arxiv.org/abs/2503.07459>

C. Patient simulators (what our simulator will build on)

16. **PATIENT-Ψ** — simulating patients for mental-health-professional training. <https://aclanthology.org/2024.emnlp-main.711/>
17. **Awesome-LLM-Patient-Simulators** — curated list, excellent starting survey. <https://github.com/FreedomIntelligence/Awesome-LLM-Patient-Simulators>
18. **AI Hospital / Agent Hospital** — Fan et al. 2024, Li et al. 2024. <https://arxiv.org/abs/2402.09742>

D. Real-world RCTs of medical chatbots (validating our effect sizes)

19. ★ **PreA trial (Nature Medicine, 2026)** — LLM chatbot for primary-to-specialist transitions; N=2,069 RCT. <https://www.nature.com/articles/s41591-025-04176-7>
20. **P&P Care (Nature Health, 2026)** — community-codesigned LLM chatbot RCT; N=2,113. <https://www.nature.com/articles/s44360-025-00021-w>
21. **PROSCA (2024)** — prostate-cancer chatbot RCT. <https://pubmed.ncbi.nlm.nih.gov/39329071/>

E. Example SMARTs (for how a protocol reads in practice)

- 22. **Pelham / ADHD SMART** — one of the clearest, most-cited SMARTs.
- 23. **Almirall, Nahum-Shani et al.** — weight-loss SMART walkthrough (same ref as #2, section on the case study).
- 24. **Oslin et al. (2008)** — alcohol dependence SMART (Murphy's motivating example).